

WORKED EXAMPLE

Adam remembers direction and squared magnitude

The same parameter has two memories and one global update counter.

m is an exponential moving average of signed gradients. v is an exponential moving average of squared gradients. A hat marks a value corrected for zero initialization.

In the momentum lessons, v meant signed parameter displacement. In this Adam lesson, v has a different meaning: squared-gradient memory. Neither m nor Adam v is the actual parameter displacement.

slide $\delta \leftrightarrow \beta_1$, $\gamma \leftrightarrow \beta_2$

Lecture 5 pages 121-122 use delta and gamma for the two decay factors. We use beta1 and beta2 to match the usual Adam notation.

Initialize m , v and the update counter once. Keep all three across batches and epochs.

WORKED EXAMPLE

Read the raw and corrected moment formulas

First average the current batch gradient, then square that result for v.

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t$$

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2$$

$$\hat{m}_t = \frac{m_t}{1 - \beta_1^t}, \quad \hat{v}_t = \frac{v_t}{1 - \beta_2^t}$$

The powers use the global update number t, starting at 1. Correction removes the moments' startup bias toward zero; this is unrelated to a network bias parameter b. The second moment is uncentered, not a variance or a Hessian.

$$\text{batch } (2, -2): \quad g = 0, \quad g^2 = 0 \neq (2^2 + (-2)^2)/2 = 4$$

WORKED EXAMPLE

Use the course denominator convention

The square-root bar includes epsilon in the lecture equation.

$$w_t = w_{t-1} - \eta \frac{\hat{m}_t}{\sqrt{\hat{v}_t + \varepsilon}}$$

Eta is still the base learning rate. Epsilon stabilizes the denominator. A common alternative places epsilon outside the root; we compare them numerically later.

$$w_0 = 1, \quad m_0 = v_0 = 0, \quad \beta_1 = 0.9, \quad \beta_2 = 0.999$$

$$\eta = 0.1, \quad g_1 = 2, \quad g_2 = -1, \quad \varepsilon = 0$$

The gradients are supplied inputs, not claimed to come from a fixed objective. Epsilon 0 is used only for these positive-denominator hand calculations. Keep full precision between steps.

WORKED EXAMPLE

update 1: raw moments

Retain the previous raw moments, not their corrected versions.

$$g_1 = 2, \quad g_1^2 = (2)^2 = 4$$

$$m_1 = 0.9(0) + 0.1(2) = 0.2$$

$$v_1 = 0.999(0) + 0.001(4) = 0.004$$

The signed gradient can reduce or reverse m . Its square is nonnegative, so opposite signs do not cancel in v .

$$\text{save raw : } (m_1, v_1) = (0.2, 0.004)$$

WORKED EXAMPLE

update 1: bias correction

Evaluate each power before subtracting from 1.

$$1 - \beta_1^1 = 1 - 0.9^1 = 0.1$$

$$1 - \beta_2^1 = 1 - 0.999^1 = 0.001$$

$$\hat{m}_1 = \frac{0.2}{0.1} \approx 2.0000000000$$

$$\hat{v}_1 = \frac{0.004}{0.001} \approx 4.0000000000$$

The two decay factors produce different correction denominators. Use both corrected values in the parameter update. Keep the raw values for the next recurrence.

WORKED EXAMPLE

update 1: denominator and parameter

d denotes the positive denominator; step is the signed quantity subtracted.

$$d = \sqrt{\hat{v}_1 + 0} \approx \sqrt{4.0000000000} \approx 2.0000000000$$

$$\text{step} = \frac{0.1\hat{m}_1}{d} \approx \frac{0.1(2.0000000000)}{2.0000000000}$$

$$\text{step} \approx 0.1000000000$$

$$w_1 \approx 1.0000000000 - (0.1000000000) = 0.9000000000$$

A negative step increases the parameter because it is subtracted. A positive step decreases it. The sign comes from the corrected first moment, not directly from the latest gradient.

WORKED EXAMPLE

update 2: raw moments

Retain the previous raw moments, not their corrected versions.

$$g_2 = -1, \quad g_2^2 = (-1)^2 = 1$$

$$m_2 = 0.9(0.2) + 0.1(-1) = 0.08$$

$$v_2 = 0.999(0.004) + 0.001(1) = 0.004996$$

The signed gradient can reduce or reverse m. Its square is nonnegative, so opposite signs do not cancel in v.

$$\text{save raw : } (m_2, v_2) = (0.08, 0.004996)$$

WORKED EXAMPLE

update 2: bias correction

Evaluate each power before subtracting from 1.

$$1 - \beta_1^2 = 1 - 0.9^2 = 0.19$$

$$1 - \beta_2^2 = 1 - 0.999^2 = 0.001999$$

$$\hat{m}_2 = \frac{0.08}{0.19} \approx 0.421052632$$

$$\hat{v}_2 = \frac{0.004996}{0.001999} \approx 2.499249625$$

The two decay factors produce different correction denominators. Use both corrected values in the parameter update. Keep the raw values for the next recurrence.

WORKED EXAMPLE

update 2: denominator and parameter

d denotes the positive denominator; step is the signed quantity subtracted.

$$d = \sqrt{\hat{v}_2 + 0} \approx \sqrt{2.499249625} \approx 1.580901523$$

$$\text{step} = \frac{0.1\hat{m}_2}{d} \approx \frac{0.1(0.421052632)}{1.580901523}$$

$$\text{step} \approx 0.026633704$$

$$w_2 \approx 0.9000000000 - (0.026633704) = 0.873366296$$

A negative step increases the parameter because it is subtracted. A positive step decreases it. The sign comes from the corrected first moment, not directly from the latest gradient.

WORKED EXAMPLE

The latest gradient changed sign, but m did not

At update 2, old positive history outweighs the new negative contribution.

$$m_2 = 0.9(0.2) + 0.1(-1) = 0.18 - 0.1 = 0.08 > 0$$

$$\hat{m}_2 = 0.08/0.19 > 0$$

$$w_2 \approx 0.9 - 0.026633704 = 0.873366296$$

The new gradient is negative, but the retained first moment is positive. Adam still subtracts a positive step in this update.

A supplied gradient sequence is enough to trace optimizer state. Without a declared objective, it does not establish that each parameter update lowers the loss.

WORKED EXAMPLE

Why dividing by 1 minus beta to the t helps

For this illustration, suppose the same scalar gradient g is repeated.

$$m_0 = 0, \quad m_1 = (1 - \beta)g$$

$$m_2 = \beta(1 - \beta)g + (1 - \beta)g = (1 - \beta^2)g$$

$$m_t = (1 - \beta^t)g \implies m_t / (1 - \beta^t) = g$$

The initial zero leaves the exponential weights summing to less than 1. Dividing by their total removes that zero-start attenuation. The same calculation applies to repeated g squared for the second moment.

For random gradients with a stationary expectation, the correction removes this initialization bias in expectation. It does not make an arbitrary changing sequence equal to its latest gradient.

WORKED EXAMPLE

Two common correction errors

Use the power of beta, not the power of its complement.

$$1 - 0.9^2 = 1 - 0.81 = 0.19$$

$$(1 - 0.9)^2 = 0.1^2 = 0.01 \quad \text{different}$$

$$\text{uncorrected first step} = 0.1(0.2)/\sqrt{0.004} \approx 0.316228$$

$$\text{corrected first step} = 0.1(2)/\sqrt{4} = 0.1$$

Omitting both corrections does not make them cancel: the first and second decay factors differ, and the second moment is under a square root.

Do not restart t at an epoch boundary. In a new training run, reset t and both moment states together.

WORKED EXAMPLE

Positive epsilon: the placement is visible

Use corrected moments $m\text{-hat}=2$ and $v\text{-hat}=4$, $\eta=.1$, $\epsilon=1$.

$$\text{course step} = \frac{0.1(2)}{\sqrt{4+1}} \approx 0.089443$$

$$\text{alternative step} = \frac{0.1(2)}{\sqrt{4+1}} \approx 0.066667$$

$$W_{old} = 1 : \quad W_{course} \approx 0.910557, \quad W_{alternative} \approx 0.933333$$

The course puts epsilon inside the root. The comparison uses a deliberately visible epsilon value, not a recommended default.

With both corrected moments zero, positive epsilon yields a zero step; epsilon 0 would make the ratio 0/0 undefined.

INDEPENDENT PRACTICE

Your turn: repeat a negative gradient

Start a separate run from the original initial state.

$$w_0 = 1, \quad m_0 = v_0 = 0, \quad (g_1, g_2) = (-2, -2)$$

$$\beta_1 = 0.9, \quad \beta_2 = 0.999, \quad \eta = 0.1, \quad \varepsilon = 0$$

For both updates, calculate raw moments, correction denominators, corrected moments, final denominator, signed step and new parameter. Work the second update using retained raw states.

Explain why the corrected second moment can equal 4 while the variance of the constant sequence $[-2, -2]$ is 0. Then identify where to add a positive epsilon under the lecture convention.

WORKED ANSWER

Practice: update 1: raw moments

Retain the previous raw moments, not their corrected versions.

$$g_1 = -2, \quad g_1^2 = (-2)^2 = 4$$

$$m_1 = 0.9(0) + 0.1(-2) = -0.2$$

$$v_1 = 0.999(0) + 0.001(4) = 0.004$$

The signed gradient can reduce or reverse m . Its square is nonnegative, so opposite signs do not cancel in v .

$$\text{save raw : } (m_1, v_1) = (-0.2, 0.004)$$

WORKED ANSWER

Practice: update 1: bias correction

Evaluate each power before subtracting from 1.

$$1 - \beta_1^1 = 1 - 0.9^1 = 0.1$$

$$1 - \beta_2^1 = 1 - 0.999^1 = 0.001$$

$$\hat{m}_1 = \frac{-0.2}{0.1} \approx -2.0000000000$$

$$\hat{v}_1 = \frac{0.004}{0.001} \approx 4.0000000000$$

The two decay factors produce different correction denominators. Use both corrected values in the parameter update. Keep the raw values for the next recurrence.

WORKED ANSWER

Practice: update 1: denominator and parameter

d denotes the positive denominator; step is the signed quantity subtracted.

$$d = \sqrt{\hat{v}_1 + 0} \approx \sqrt{4.0000000000} \approx 2.0000000000$$

$$\text{step} = \frac{0.1\hat{m}_1}{d} \approx \frac{0.1(-2.0000000000)}{2.0000000000}$$

$$\text{step} \approx -0.1000000000$$

$$w_1 \approx 1.0000000000 - (-0.1000000000) = 1.1000000000$$

A negative step increases the parameter because it is subtracted. A positive step decreases it. The sign comes from the corrected first moment, not directly from the latest gradient.

WORKED ANSWER

Practice: update 2: raw moments

Retain the previous raw moments, not their corrected versions.

$$g_2 = -2, \quad g_2^2 = (-2)^2 = 4$$

$$m_2 = 0.9(-0.2) + 0.1(-2) = -0.38$$

$$v_2 = 0.999(0.004) + 0.001(4) = 0.007996$$

The signed gradient can reduce or reverse m . Its square is nonnegative, so opposite signs do not cancel in v .

$$\text{save raw : } (m_2, v_2) = (-0.38, 0.007996)$$

WORKED ANSWER

Practice: update 2: bias correction

Evaluate each power before subtracting from 1.

$$1 - \beta_1^2 = 1 - 0.9^2 = 0.19$$

$$1 - \beta_2^2 = 1 - 0.999^2 = 0.001999$$

$$\hat{m}_2 = \frac{-0.38}{0.19} \approx -2.0000000000$$

$$\hat{v}_2 = \frac{0.007996}{0.001999} \approx 4.0000000000$$

The two decay factors produce different correction denominators. Use both corrected values in the parameter update. Keep the raw values for the next recurrence.

WORKED ANSWER

Practice: update 2: denominator and parameter

d denotes the positive denominator; step is the signed quantity subtracted.

$$d = \sqrt{\hat{v}_2 + 0} \approx \sqrt{4.0000000000} \approx 2.0000000000$$

$$\text{step} = \frac{0.1\hat{m}_2}{d} \approx \frac{0.1(-2.0000000000)}{2.0000000000}$$

$$\text{step} \approx -0.1000000000$$

$$w_2 \approx 1.1000000000 - (-0.1000000000) = 1.2000000000$$

A negative step increases the parameter because it is subtracted. A positive step decreases it. The sign comes from the corrected first moment, not directly from the latest gradient.

WORKED ANSWER

Practice checkpoint: magnitude is not variability

The two repeated negative gradients are a useful bias-correction check.

$$m_1 = -0.2, \quad v_1 = 0.004, \quad m_2 = -0.38, \quad v_2 = 0.007996$$

$$\hat{m}_1 = \hat{m}_2 = -2, \quad \hat{v}_1 = \hat{v}_2 = 4$$

$$w_1 = 1 - (-0.1) = 1.1, \quad w_2 = 1.1 - (-0.1) = 1.2$$

$$\text{variance} = ((-2 - (-2))^2 + (-2 - (-2))^2)/2 = 0$$

The squared-gradient level is 4, but deviation around its constant mean is 0. Adam uses the former. With positive epsilon, its course denominator is $\sqrt{v\text{-hat} + \text{epsilon}}$.